Smart Gate Pass Management System

Minor Project Presentation | Bachelor of Fashion Technology

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Introduction

- SAPL sends materials (fabric, trims, garments) outside the factory for washing, dyeing, printing and embroidery.
- Every outward movement of material needs a gate pass to track approvals and returns.
- Earlier, this was a fully paper-based process — handwritten slips, manual signatures, physical filing.
- This project digitises the entire process using a Flutter mobile app with a Firebase backend.
- Four user roles are supported: Employee, Supervisor, Security Guard, and Admin.

Problem Statement

- Paper gate pass slips were easily lost, damaged, or misplaced.
- No easy way to check whether material sent out had actually returned.
- Supervisors had to be physically present to approve every request, causing delays.
- Searching old records for audits meant going through stacks of paper files.
- Security had no reliable way to verify a gate pass at the gate.
- Generating reports required manual counting and cross-checking.

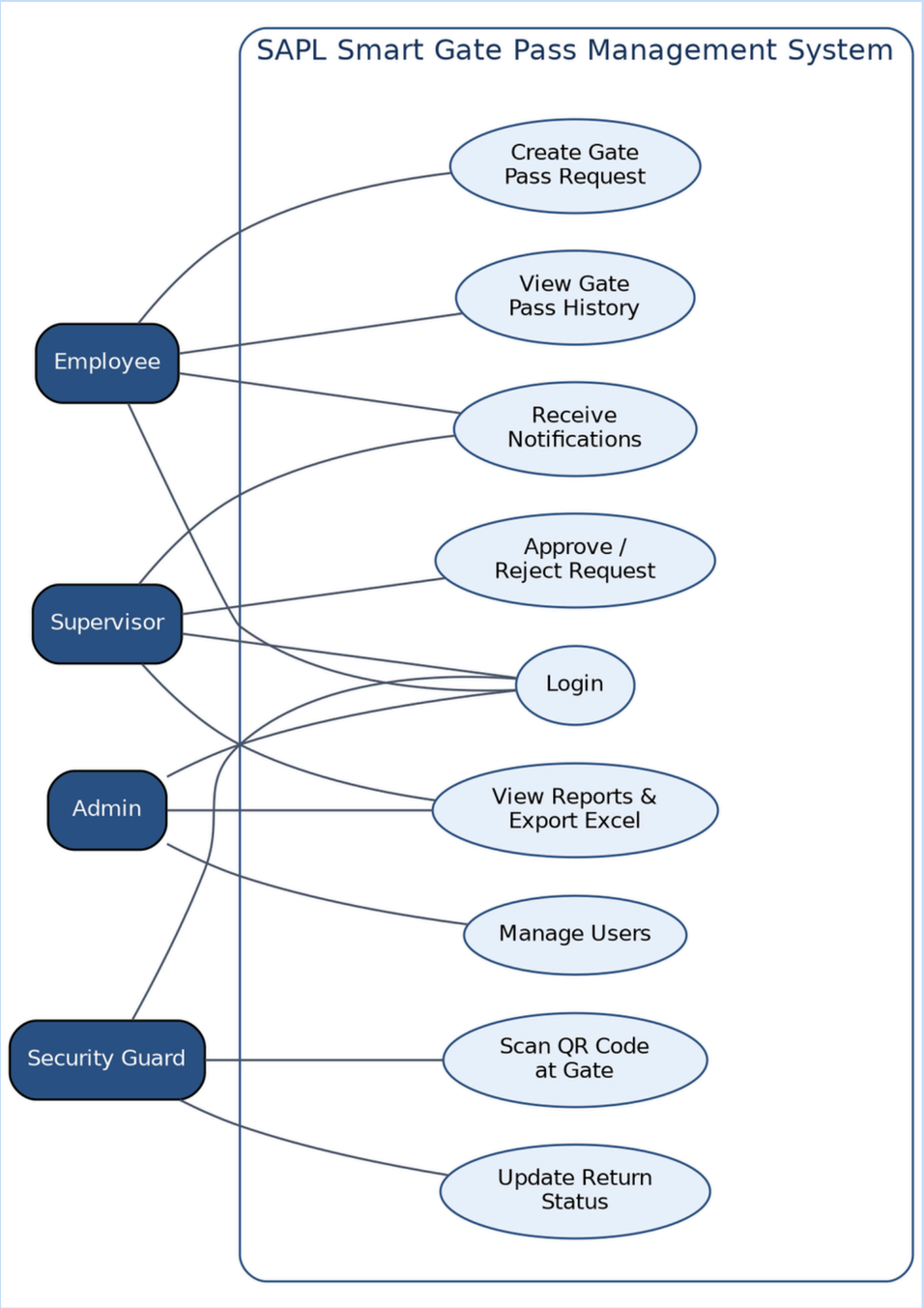
Objectives

- Replace the manual gate pass process with a digital mobile application.
- Let employees raise gate pass requests directly from their phones.
- Allow supervisors to approve or reject requests remotely.
- Generate a QR code for every approved pass for quick security verification.
- Track material returns and flag overdue items.
- Provide reports that can be exported to Excel for record-keeping.

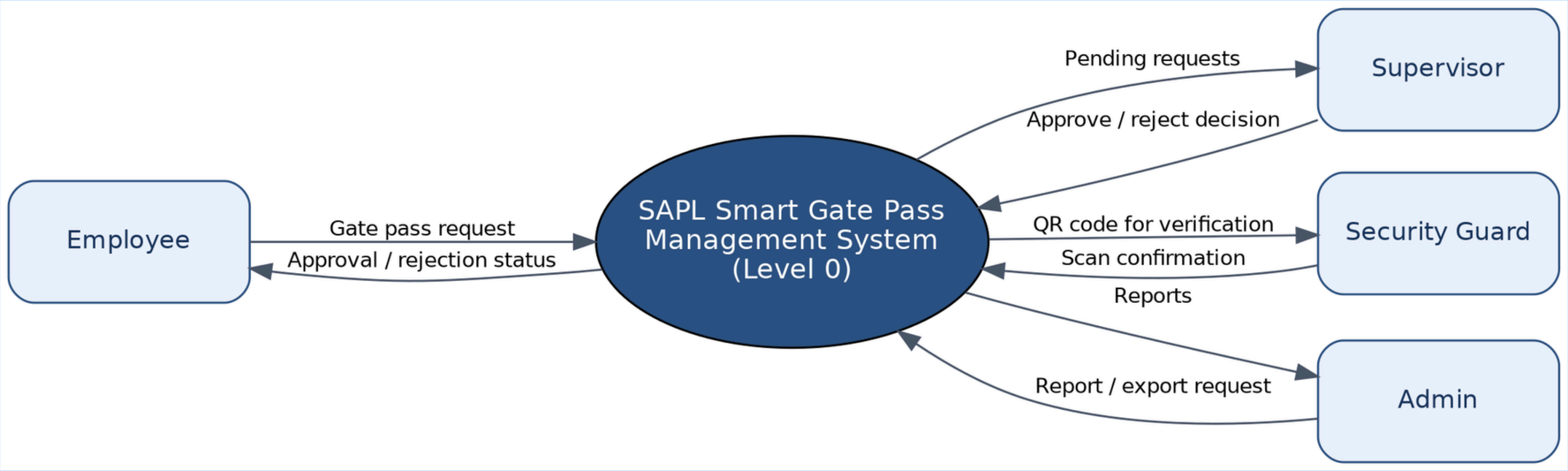
Methodology — Observation Method

- Spent time on the factory floor and at the security gate to study the real paper-based process.
- Observed employees filling gate pass slips and noted the fields they commonly filled in.
- Observed supervisors' approval process and the delays it caused
- Observed the security guard's verification process at the gate.
- Held informal conversations with staff to identify pain points.
- Used these findings to design the app's fields, screens and workflow.

System Design — Use Case Diagram



Data Flow Diagram



Main Modules

- Employee Login
- Employee Dashboard
- Create Gate Pass
- Gate Pass History
- Supervisor Dashboard
- Approval / Rejection
- QR Code Generation
- Security QR Scanner
- Return Tracking
- Reports Dashboard
- Excel Export
- In-app Notifications

Workflow

1. Employee logs into the application.

2. Employee creates a gate pass with material, vendor, vehicle and return date details.

3. Request is stored in Firebase Firestore.

4. Supervisor reviews and approves or rejects the request.

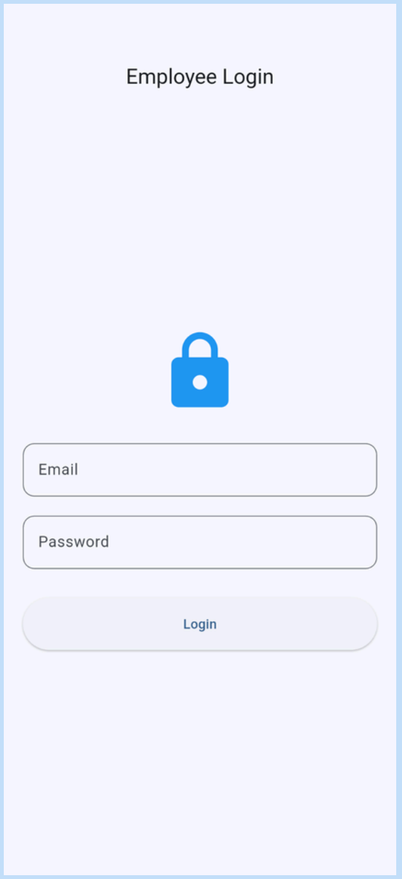
5. On approval, a QR code is generated.

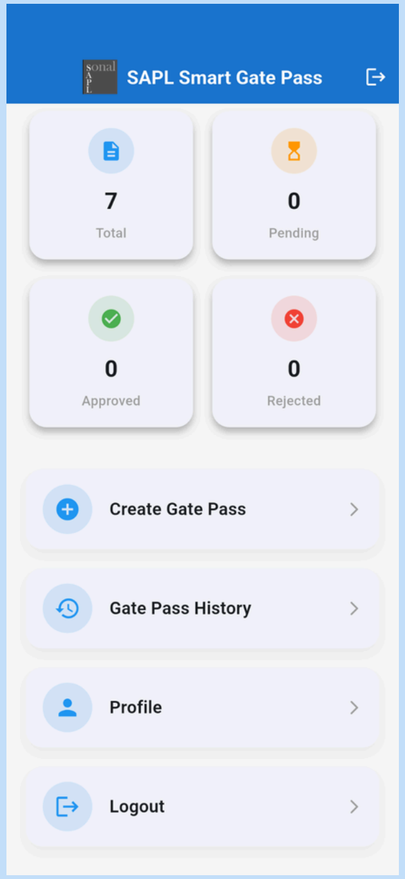
6. Security scans the QR code when material leaves or returns.

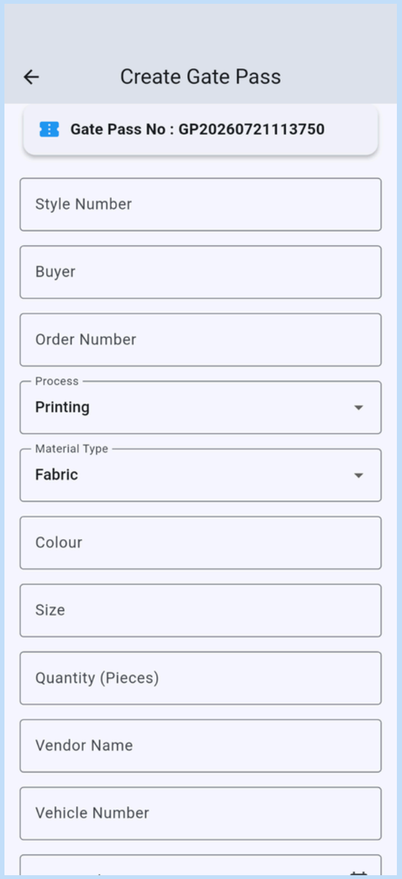
7. Return status is updated in the system.

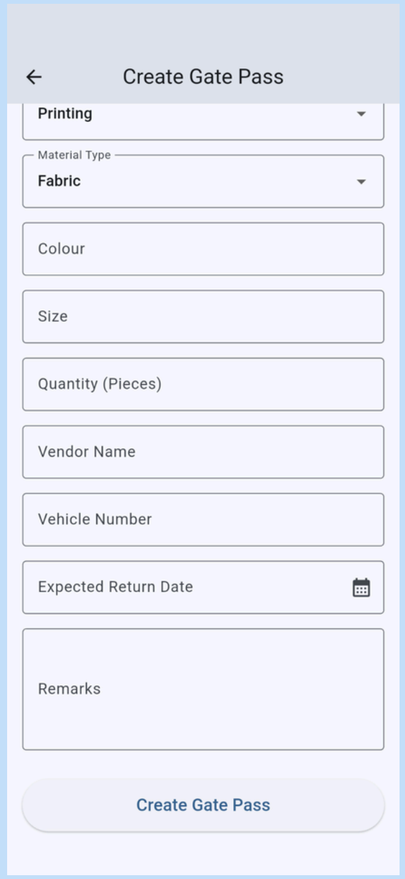
8. Supervisor / Admin views reports and exports data to Excel.

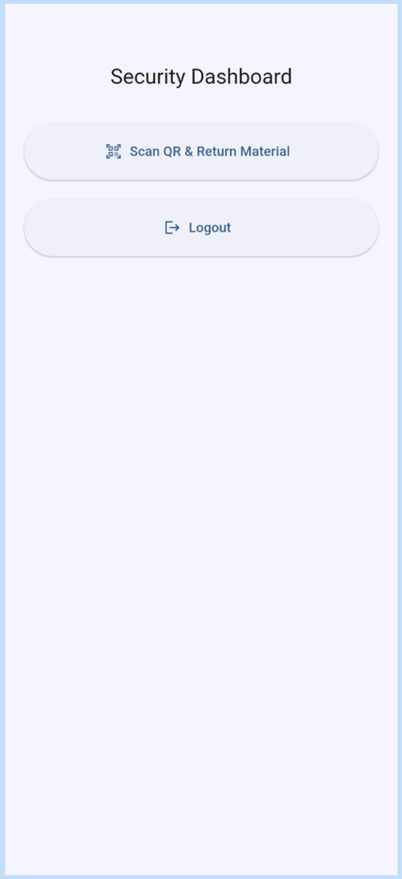
App Screenshots (Employee & Security Login)



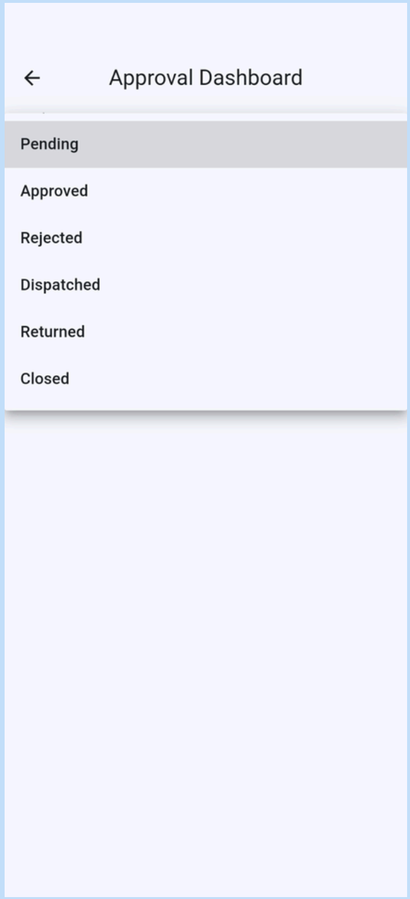


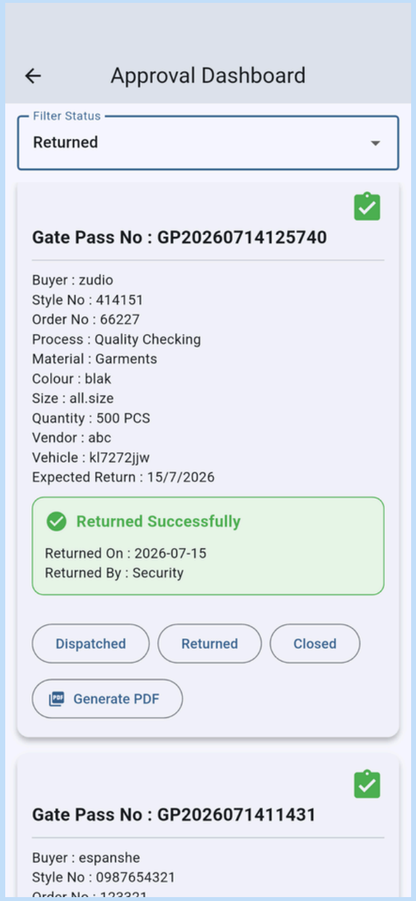


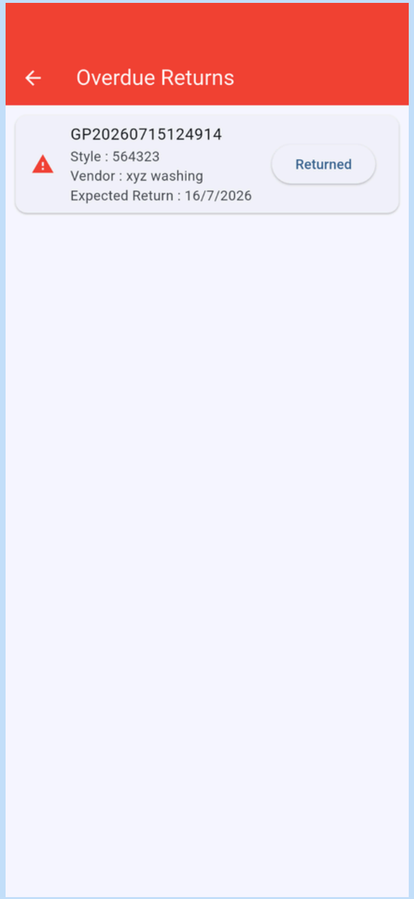


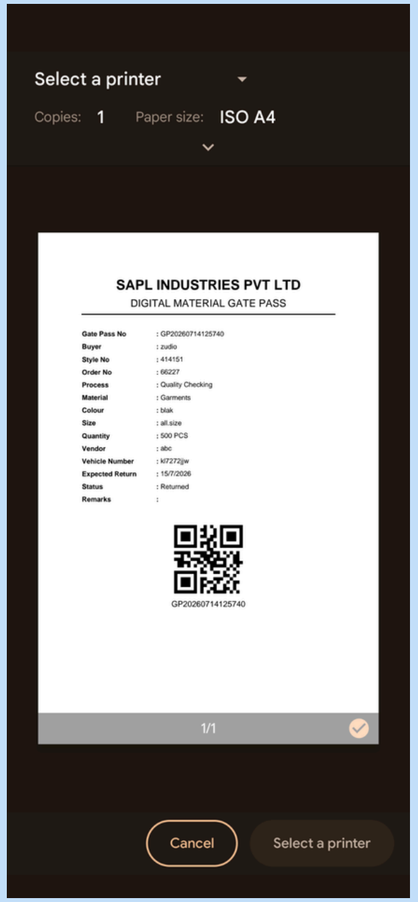


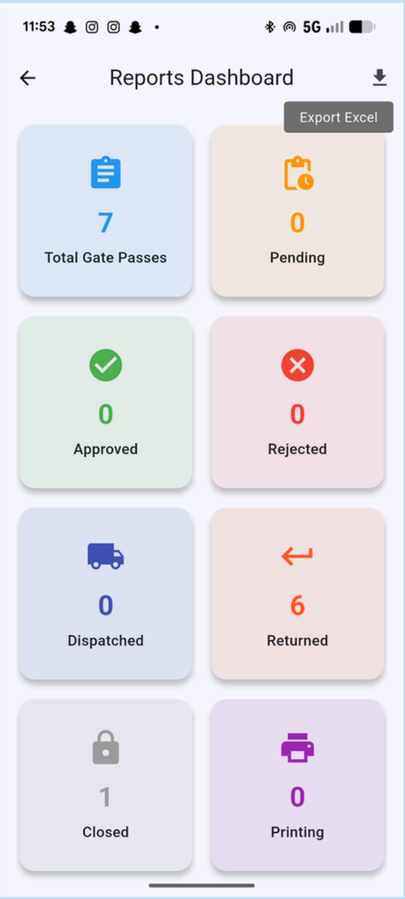
App Screenshots (Supervisor Login)











Results

- Gate pass requests could be created in under a minute, versus filling a paper slip.
- Supervisors approved or rejected requests remotely without delays.
- QR codes were scanned and verified successfully at the security module.
- Overdue / pending returns were correctly flagged by the system.
- Reports were filtered and exported to Excel with ease.

Limitations & Future Scope

Limitations

- Depends on an active internet connection (Firebase real-time).
- No integration yet with SAPL's ERP / SAP systems.
- No barcode scanning for individual material items.
- Overdue-return reminders are basic.

Future Scope

- SAP / ERP integration for seamless data flow.
- Barcode / QR scanning for individual materials.
- Smarter automated reminders for overdue returns.
- Detailed analytics dashboard + offline support.

Conclusion

The SAPL Smart Gate Pass Management System replaces a manual, paper-based process with a fast, transparent and trackable digital workflow — from request, to approval, to QR-based verification, to return tracking and reporting. It also gave hands-on exposure to real-world app development and system design (DFDs, ER diagrams, architecture) grounded in an actual industry process.

Thank You